

VLSI phase Report

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1. Design Compiler part

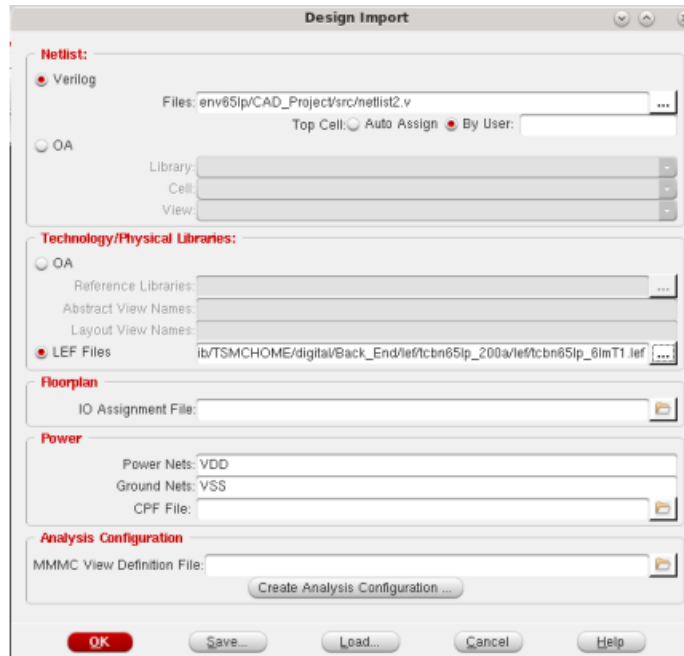
Our main module is **CORDIC_ip**, which includes a submodule named **CORDIC_element**. First, we compile the submodule in **Design Compiler**, and then we compile the main module.

The compilation was performed several times with different clock cycles, and the results were recorded. (the execution commands in Design Compiler are saved in a file called commands.tcl.txt).

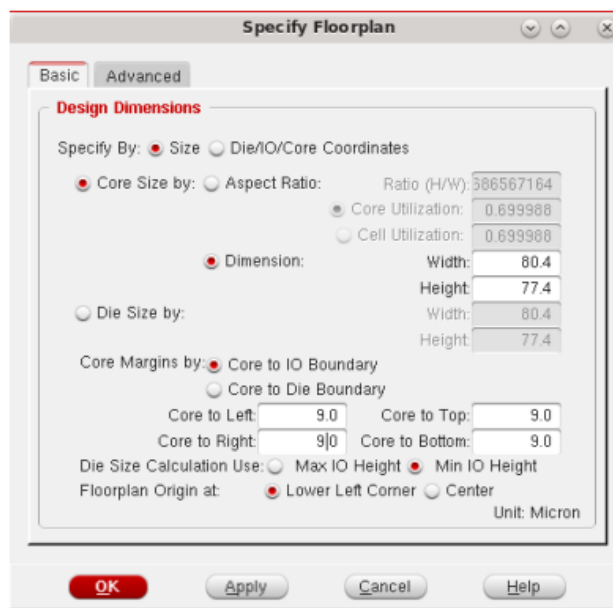
clock	power	Area	Energy
5ns	0.7832	2899	3.916
2ns	1.9354	2809	3.8708
1ns	3.8710	2809	3.871
0.8ns	4.8363	2809	3.869
0.6ns	6.3295	3010	3.7977
0.5ns	7.7080	3043	3.854
0.4ns	9.8118	3267	3.9247
0.37ns	10.8311	3335.76	4
0.35ns	12.2978	3590	4.3

2. Innovus part

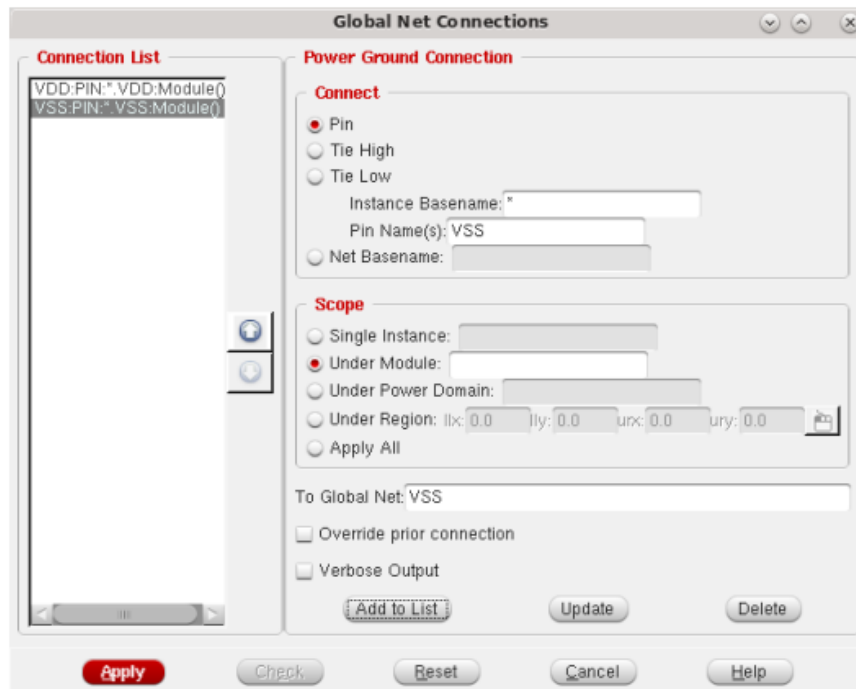
- Import netlist



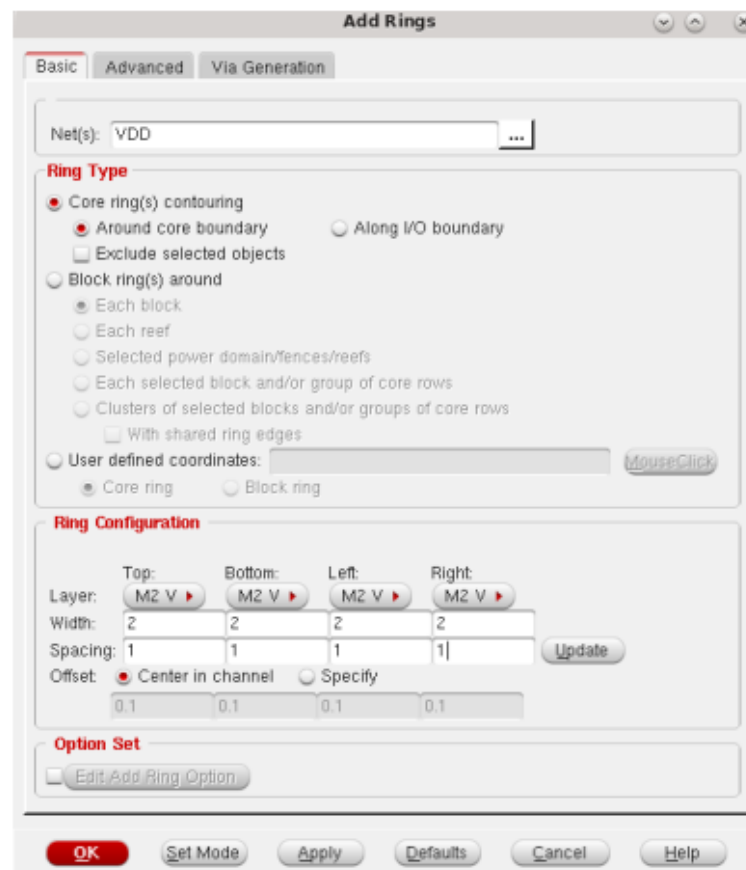
- Specify floorplan

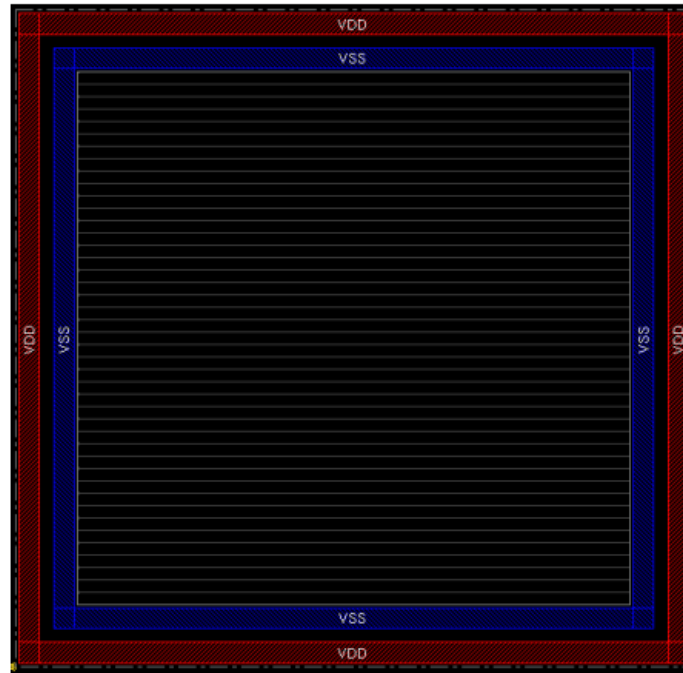


- Define global net connections for VDD and VSS.



- Add rings.





- Connect power strips.

Add Stripes

Basic Advanced Via Generation

Set Configuration

Net(s): VDD ...

Layer: M2 Direction: ☒ Vertical ☐ Horizontal

Width: 2 Spacing: 2

Set Pattern

☐ Set-to-set distance: 100

☒ Number of sets: 6

☐ Bumps ☒ Over ☐ Between

☐ Over P/G pins Pin layer: Top pin layer ☐ Max pin width: 0

☒ Master name: ☐ Selected blocks ☐ All blocks

Stripe Boundary

☒ Core ring

☐ Pad ring ☐ Inner ☒ Outer

☐ Design boundary ☒ Create pins

☐ Each selected block/domain/fence

☐ All domains

☐ Specify rectangular area

☐ Specify rectilinear area

First/Last Stripe

Start from: ☒ left ☐ right

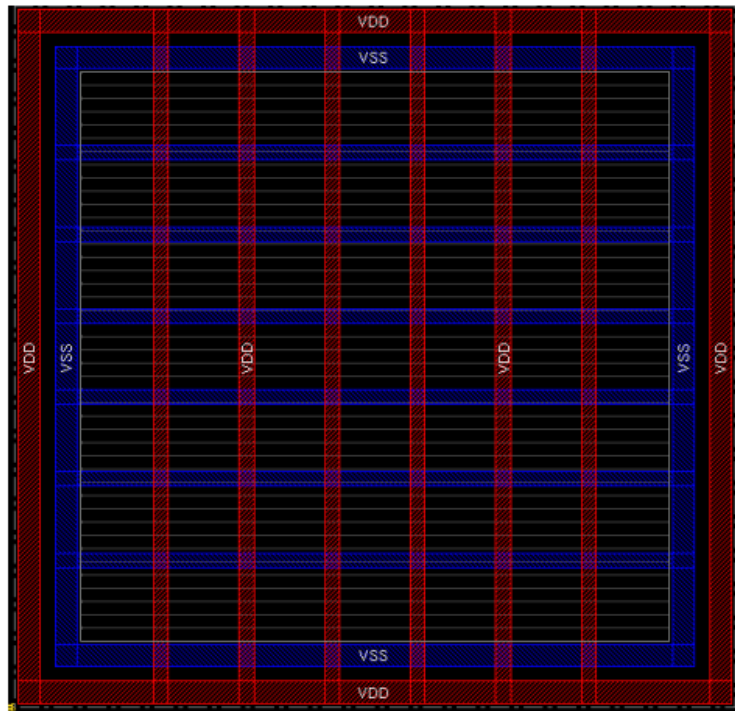
☒ Relative from core or selected area

X from left: 10 X from right: 10

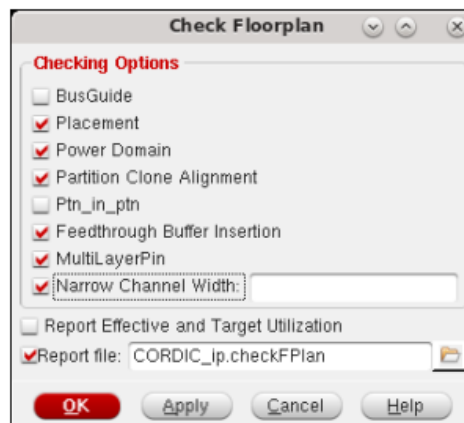
☐ Absolute locations

Option Set

☐ Edit Add Stripe Option



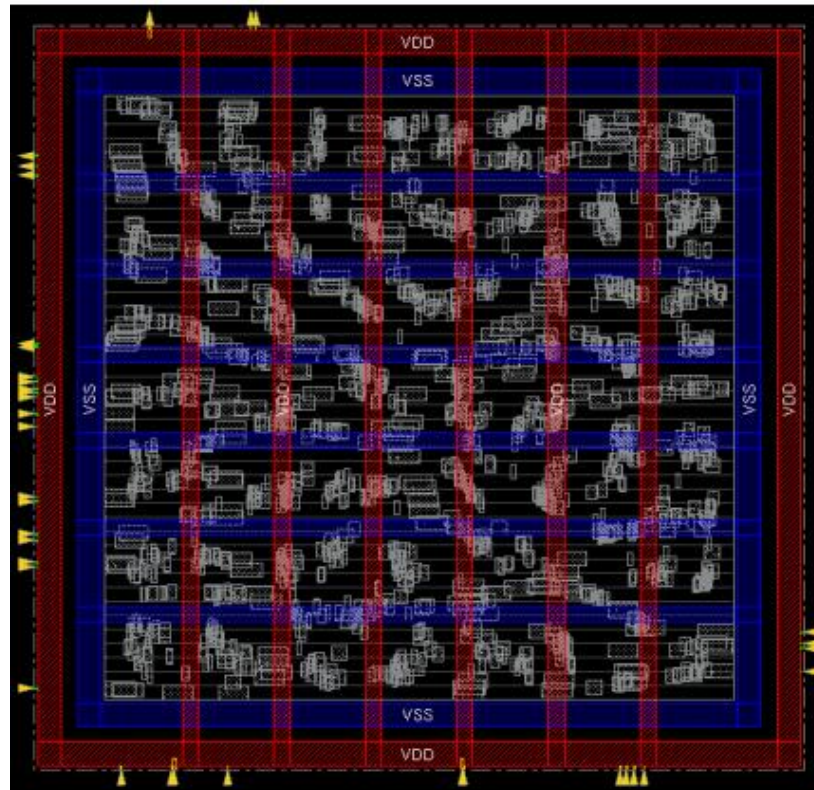
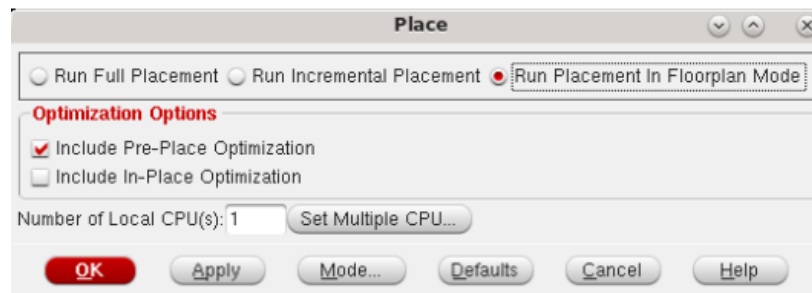
- Check floorplan



```
Checking Preroutes.....
No. of regular pre-routes not on tracks : 0
Calling CheckPlace .....
Begin checking placement ... (start mem=628.8M, init mem=630.8M)
*info: Placed = 0
*info: Unplaced = 1802
Placement Density:70.00%(4356/6223)
Finished checkPlace (cpu: total=0:00:00.2, vio checks=0:00:00.0; mem=630.8M)
Calling VerifyPowerDomain .....
```

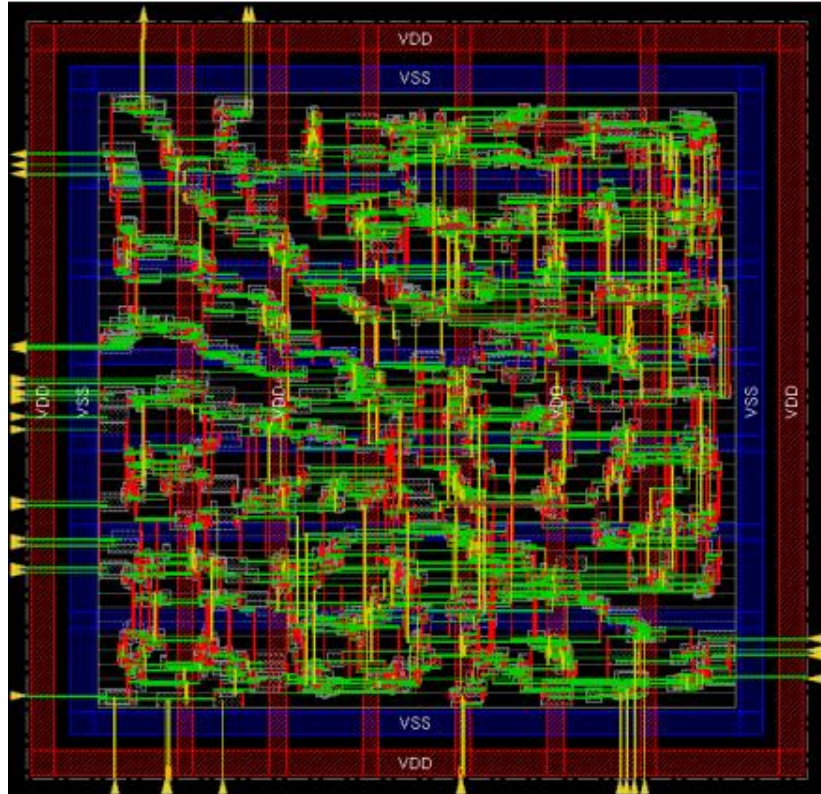
about **70% of the chip area** is occupied.

- Placement



- Routing





Congestion distribution:

Remain	cntH		cntV	
-5:	0	0.00%	24	0.61%
-4:	0	0.00%	8	0.20%
-3:	0	0.00%	11	0.28%
-2:	0	0.00%	14	0.35%
-1:	0	0.00%	24	0.61%
0:	0	0.00%	37	0.93%
1:	0	0.00%	45	1.14%
2:	0	0.00%	69	1.74%
3:	2	0.05%	76	1.92%
4:	2	0.05%	149	3.76%
5:	3956	99.90%	3503	88.46%

Total length: 1.216e+04um, number of vias: 10087
 M1(H) length: 5.363e+02um, number of vias: 5422
 M2(V) length: 4.817e+03um, number of vias: 3849
 M3(H) length: 5.244e+03um, number of vias: 727
 M4(V) length: 1.320e+03um, number of vias: 84
 M5(H) length: 2.360e+02um, number of vias: 5
 M6(V) length: 5.800e+00um, number of vias: 0
 M7(H) length: 0.000e+00um

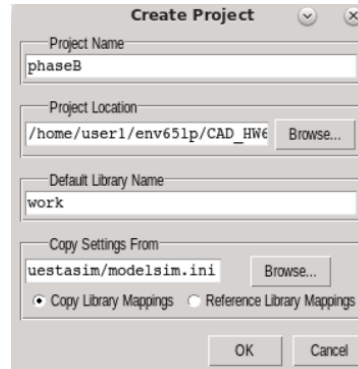
Power Consumption:

```
innovus 1>
innovus 1> report area
Depth  Name          #Inst  Area (um^2)
-----
0      CORDIC ip      1217   3590.28
1
innovus 2> █
```

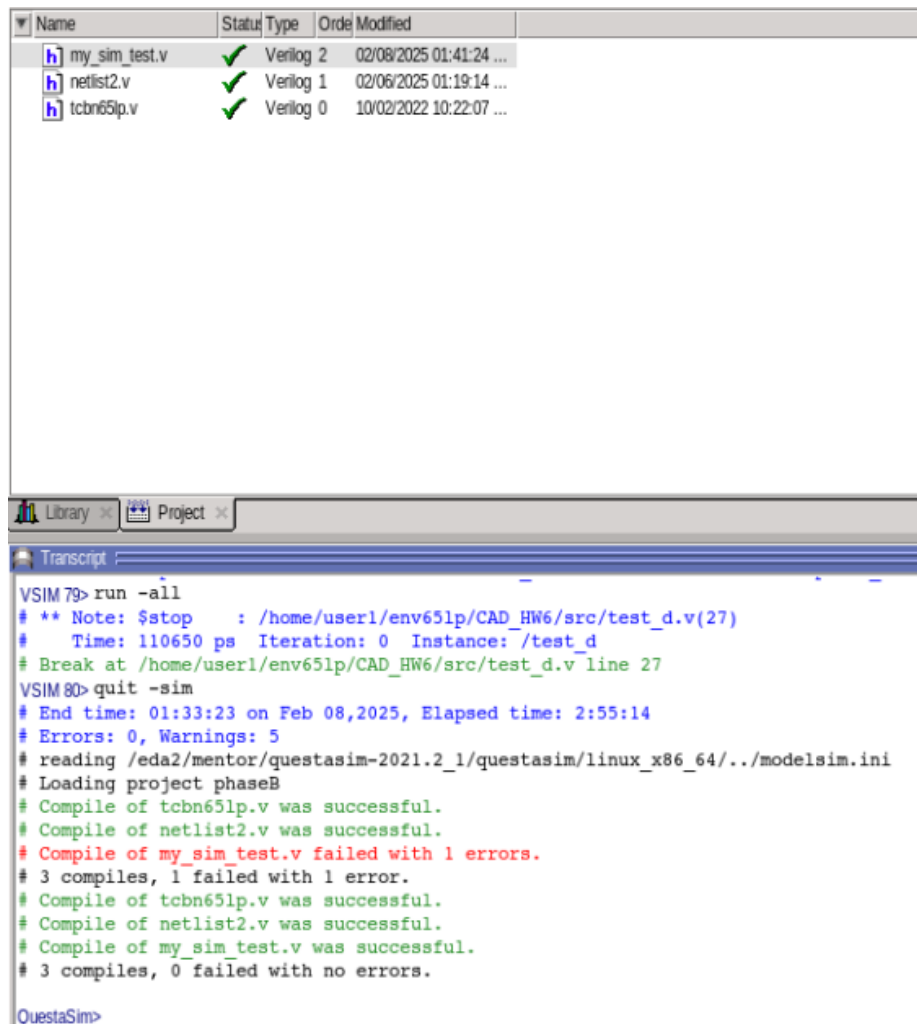
The power reported by Innovus is approximately equal to the power measured by Design Compiler.

3. Questasim part

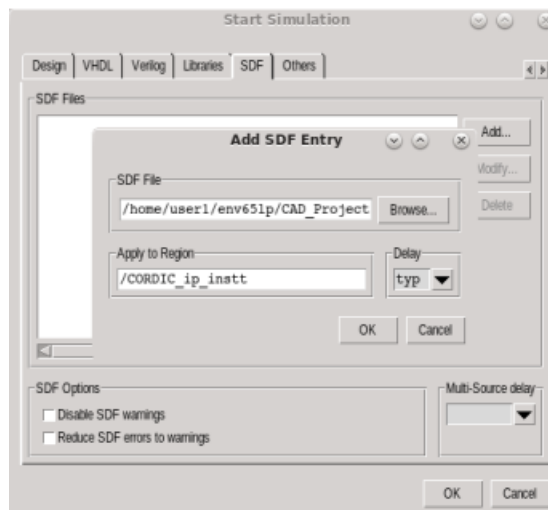
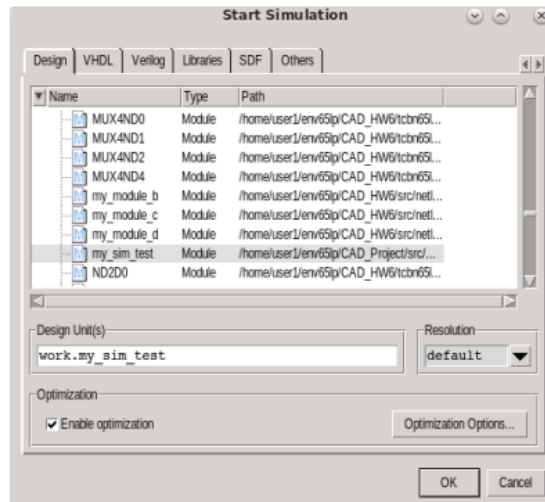
After opening the software, we create a new project.



Then we add three files to the project — the netlist, the technology file, and the testbench — and compile them.



We also add the SDF file for timing simulation.



In the testbench (my_sim.v), MATLAB-generated data are read from two files.

The module outputs are stored in another file.

A MATLAB script compares them to verify the accuracy percentage.

In the testbench, 1 clock cycle = 1 nanosecond, which matches the post-synthesis timing in Design Compiler.

The MATLAB verification confirmed the correctness of the synthesized module.

The module was tested with 1000 different input data.

Correct Results:
1000

